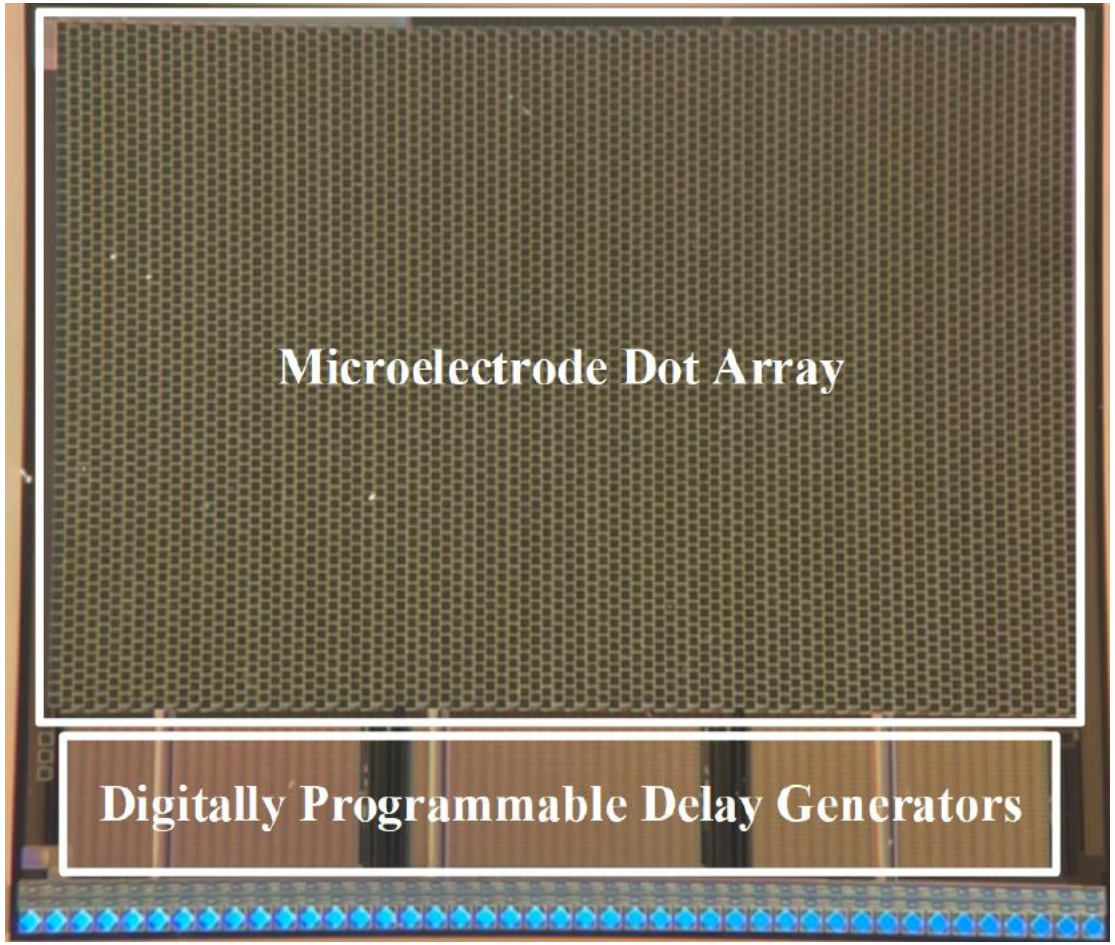
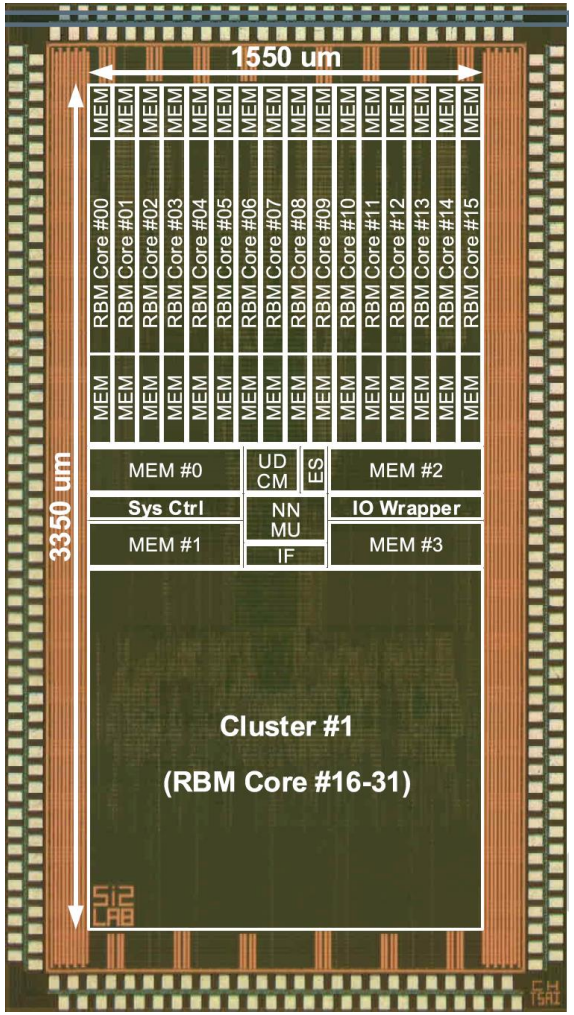
Fast Medical Tests

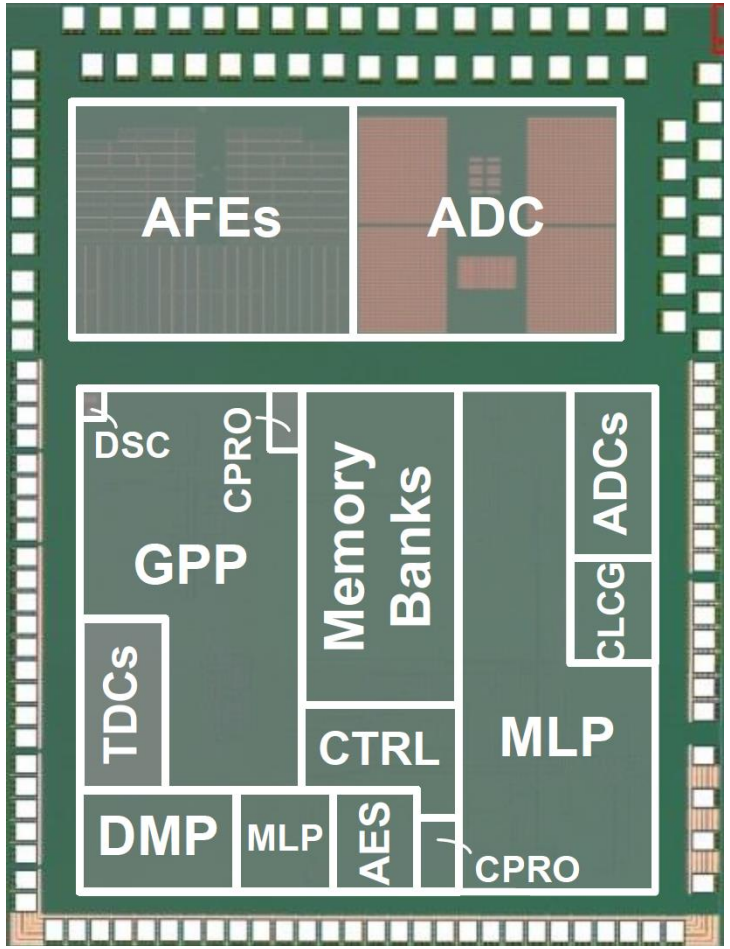
- A pattern-based digital microfluidic lab-on-CMOS chip is presented for nucleic acid amplification test.
- Yun-Sheng Chan and **Chen-Yi Lee**, “A Programmable Bio-Chip with Adaptive Pattern-Control Micro-Electrode-Dot-Array”, IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 69, no. 11, pp. 4513-4517, Nov. 2022, doi: 10.1109/TCSII.2022.3184749.



Edge AI

- An energy-efficient restricted Boltzmann machine (RBM) processor (RBM-P) supporting on-chip learning and inference is proposed for machine learning and Internet of Things (IoT) applications
- Chang-Hong Tsai, Wan-Ju Yu, Wing H. Wong, and **Chen-Yi Lee**, “A 41.3pJ/26.7pJ per Neuron Weight RBM Processor Supporting On-Chip Learning/Inference for IoT Applications”, IEEE Trans. On JSSC, Vol. 52, Issue 10, Oct. 2017, PP. 2601-2612.



Mobile Healthcare

- An ECG-SoC Chip with embedded Machine-Learning Models For real-time Heart Disease Monitoring
- Shu-Yu Hsu, Yingchieh Ho, Po-Yao Chang, Chauchin Su, and **Chen-Yi Lee**, “A 48.6-to-105.2 μW Machine Learning Assisted Cardiac Sensor SoC for Mobile Healthcare Applications”, IEEE Journal of Solid-State Circuits, Vol. 49, No. 4, April 2014, PP. 801-811.